EFY TESTED
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UART Automatic BAUD RATE DETECTOR

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While tinkering with electronics we often come across a UART device whose baud rate we do not know but need to know. (UART stands for Universal Asynchronous Receiver/Transmitter, which is a microchip with programming that controls a computer's interface to its attached serial devices.) Connecting the UART device to a serial monitor and checking for different baud rates is a time consuming process. So, here is a simple Arduino based proposed circuit that can be used to find the baud rate of the UART device. The parts needed to build it are listed under the bill of material table.

TABLE 1
BILL OF MATERIAL

Items	Description
Arduino Uno (1)	Main board for checking baud rate
Jumper wire (2)	For connection
Serial device (1)	Any serial device with unknown baud rate

TABLE 2
STANDARD BAUD RATES

Time	Baud Rate
3333μs (3.3ms)	300
833μs	1200
416μs	2400
208μs	4800
104μs	9600
69μs	14400
52μs	19200
34μs	28800
26μs	38400
17.3μs	57600
8μs	115200

Circuit diagram of the UART automatic baud rate detector is shown in Fig. 1. The circuit built around Arduino Uno is simple. The UART device's GND is connected to GND of Arduino Uno and TX pin of UART device is connected to RX (0th pin - D0) of Arduino Uno.

Install the Arduino IDE and write the source code for this project and save it as auto_baud_detector.ino. A screenshot of the source code is shown in Fig. 2.

Connect the Arduino Uno board to a laptop/desktop using a USB cable and upload the source code by selectting the right board and port name. Once the Arduino sketch auto_baud_detector.ino is uploaded, open the Arduino serial monitor or any other device.

For baud rate detection, all you need is to measure the shortest interval between two flanks (rising→falling or falling→rising) for some amount of traffic. Assuming the communication wiring is good, the shortest duration between two flanks will be the duration of a single bit. The baud rate will then be 1 divided by that duration (in seconds). You would probably want to round that figure to the closest standard baud rate (see the table for standard baud rates), once measured.

The code prepared for the baud rate detector device attempts to establish communication with the serial device whose baud rate is unknown to us. It tries out different baud rates and establishes a connection with the serial port.

Testing

Once the Arduino sketch is uploaded, open the Arduino serial monitor or any other serial console with 9600 baud rate. Connect TX pin of the unknown UART device to RX pin of Arduino Uno, as shown in Fig. 2. It will display the nearest standard baud rate. The screenshot of serial console application (putty) in Fig. 3 shows one of the test results.

This baud rate detector can now be used to find out the baud rate of

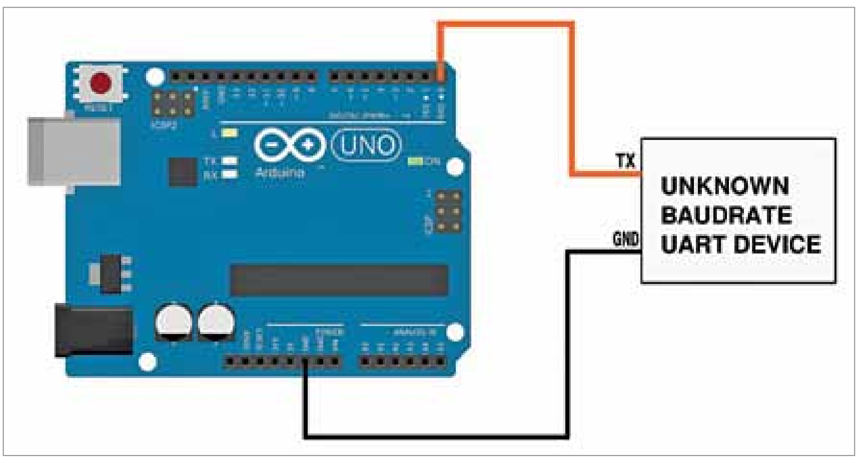

Fig. 1: Circuit diagram